Grade 1 Mathematics

Balochistan Textbook Board

Solved Exercises — Complete Book

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Unit 1 — Whole Numbers (pages 5 to 36)

Students' learning outcomes (page 5): After completing this unit, students will be able to: count objects and numbers to and across 99 (2 digit numbers) forwards and backwards, beginning from 0, 1 or from any given number; read and write numbers up to 99 (2 digit numbers) in numerals and words; recognize the place value of each digit in 2 digit numbers (tens, ones / units); compare and order numbers up to 99 using appropriate language (for instance: more and less, greater, smaller, equal to, same as, increasing, decreasing, smallest to largest and vice versa etc.); identify numbers before, after and between two numbers; recognize the position of objects and write it using ordinal numbers up to 10.

Count Numbers 1 to 9 (page 6) — Let's count and read numbers

Count the pigeons — number

1 pigeon — 1

2 pigeons — 2

3 pigeons — 3

4 pigeons — 4

5 pigeons — 5

6 pigeons — 6

7 pigeons — 7

8 pigeons — 8

9 pigeons — 9

Learning Task (page 7) — Count the objects and encircle the correct number

How many watermelons? (options: 1, 2, 5)

✓ 1

2

5

How many leaves? (options: 1, 2, 3)

1

✓ 2

3

How many cups? (options: 1, 3, 4)

1

✓ 3

4

How many butterflies? (options: 4, 5, 6)

4

✓ 5

6

How many clocks? (options: 6, 7, 8)

6

7

✓ 8

How many flowers? (options: 2, 3, 4)

2

3

✓ 4

How many horses? (options: 6, 8, 9)

✓ 6

8

9

Key point (page 7): 9 is the greatest 1-digit number.

Number Zero (0) (page 8)

Counting starts from zero. Zero is the smallest whole number.

Q: How many cars are there?

A: There are 3 cars on the plate.

Q: How many apples are there?

A: There are 2 apples on the plate.

Q: How many candies are there?

A: There is 1 candy on the plate.

Q: How many cups are there?

A: There are 0 cups — the plate is empty, and empty means zero.

Learning Task 1 (page 9) — Match the numbers with objects

Objects — number

4 mangoes — 4

Empty box (nothing in it) — 0

7 birds — 7

9 stars — 9

3 ice creams — 3

2 chairs — 2

6 apples — 6

1 kitten — 1

8 candies — 8

5 basketballs — 5

Number 0 to 10 (page 10) — Count and read

Number — word

0 — Zero

1 — One

2 — Two

3 — Three

4 — Four

5 — Five

6 — Six

7 — Seven

8 — Eight

9 — Nine

10 — Ten

Numbers 11 to 20 (page 11) — Count and read

Number — word

11 — Eleven

12 — Twelve

13 — Thirteen

14 — Fourteen

15 — Fifteen

16 — Sixteen

17 — Seventeen

18 — Eighteen

19 — Nineteen

20 — Twenty

Numbers 21 to 30 (page 12) — Count and read

Number — word

21 — Twenty One

22 — Twenty Two

23 — Twenty Three

24 — Twenty Four

25 — Twenty Five

26 — Twenty Six

27 — Twenty Seven

28 — Twenty Eight

29 — Twenty Nine

30 — Thirty

Numbers 31 to 40 (page 13) — Count and read

Number — word

31 — Thirty one

32 — Thirty two

33 — Thirty three

34 — Thirty four

35 — Thirty five

36 — Thirty six

37 — Thirty seven

38 — Thirty eight

39 — Thirty nine

40 — Forty

Numbers 41 to 50 (page 14) — Count and read

Number — word

41 — Forty One

42 — Forty Two

43 — Forty Three

44 — Forty Four

45 — Forty Five

46 — Forty Six

47 — Forty Seven

48 — Forty Eight

49 — Forty Nine

50 — Fifty

Numbers 51 to 60 (page 15) — Count and read

Number — word

51 — Fifty One

52 — Fifty Two

53 — Fifty Three

54 — Fifty Four

55 — Fifty Five

56 — Fifty Six

57 — Fifty Seven

58 — Fifty Eight

59 — Fifty Nine

60 — Sixty

Numbers 61 to 70 (page 16) — Count and read

Number — word

61 — Sixty One

62 — Sixty Two

63 — Sixty Three

64 — Sixty Four

65 — Sixty Five

66 — Sixty Six

67 — Sixty Seven

68 — Sixty Eight

69 — Sixty Nine

70 — Seventy

Numbers 71 to 80 (page 17) — Count and read

Number — word

71 — Seventy One

72 — Seventy Two

73 — Seventy Three

74 — Seventy Four

75 — Seventy Five

76 — Seventy Six

77 — Seventy Seven

78 — Seventy Eight

79 — Seventy Nine

80 — Eighty

Numbers 81 to 90 (page 18) — Count and read

Number — word

81 — Eighty One

82 — Eighty Two

83 — Eighty Three

84 — Eighty Four

85 — Eighty Five

86 — Eighty Six

87 — Eighty Seven

88 — Eighty Eight

89 — Eighty Nine

90 — Ninety

Numbers 91 to 99 (page 19) — Count and read

Number — word

91 — Ninety One

92 — Ninety Two

93 — Ninety Three

94 — Ninety Four

95 — Ninety Five

96 — Ninety Six

97 — Ninety Seven

98 — Ninety Eight

99 — Ninety Nine

Numbers up to 99 (Forward) (page 20) — Count numbers

Count forward from 1 to 99 in order: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99. (Trace and read the flower chart in the book.)

Numbers up to 99 (Backward) (page 21) — Count numbers backward

Count backward from 99 down to 1 in order: 99, 98, 97, 96, 95, 94, 93, 92, 91, 90, 89, 88, 87, 86, 85, 84, 83, 82, 81, 80, 79, 78, 77, 76, 75, 74, 73, 72, 71, 70, 69, 68, 67, 66, 65, 64, 63, 62, 61, 60, 59, 58, 57, 56, 55, 54, 53, 52, 51, 50, 49, 48, 47, 46, 45, 44, 43, 42, 41, 40, 39, 38, 37, 36, 35, 34, 33, 32, 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1. (Trace and read the

flower chart in the book.)

Place Value (page 22)

Place value tells us the position / place of a digit in a number. Value of a digit depends on its place in a number. Example: Sara has 9 pencils, she got one more. Now she has 10 pencils. One bundle of 10 pencils means One ten. 10 ones make 1 ten.

Place value example — Sara's pencils: 9 + 1 = ?

Step 1: Sara has 9 pencils.

Step 2: She gets 1 more pencil.

Step 3: 9 + 1 = 10.

Step 4: 10 pencils make one bundle of 10, which is 1 ten.

Answer: 10 pencils = 1 ten

Key points (page 22): 10 ones make 1 ten. 10 is the smallest 2 digit number.

Position of a digit (page 23)

In number 10, the place value of 0 is 0 ones and the place value of 1 is 1 ten. Example — count the beads together: the beads group into 10 and 3. Now these are 1 group of 10 and one group of 3. Total number of beads = 13. The place value of 1 is 1 ten. The place value of 3 is 3 ones. (Note: the book prints '1 ones' here by mistake; the correct value is 3 ones, since there are 3 single beads.) Example — place value of 35: the Tens box holds 3 ten-strips and the Ones box holds 5 single squares. The place value of 3 is tens and 5 is ones: 35 = 3 tens + 5 ones.

Learning Tasks 1 (page 24) — Count tens and ones and write the number

(1) 2 ten-strips and 4 ones

Step 1: Count the ten-strips: there are 2 strips of 10.

Step 2: 2 tens = 20.

Step 3: Count the single squares: there are 4 ones.

Step 4: 20 + 4 = 24.

Answer: 24 — 2 Tens, 4 Ones

(2) 3 ten-strips and 6 ones

Step 1: Count the ten-strips: there are 3 strips of 10.

Step 2: 3 tens = 30.

Step 3: Count the single squares: 1 + 2 + 3 = 6 ones.

Step 4: 30 + 6 = 36.

Answer: 36 — 3 Tens, 6 Ones

(3) 4 ten-strips and 6 ones

Step 1: Count the ten-strips: there are 4 strips of 10.

Step 2: 4 tens = 40.

Step 3: Count the single squares: 2 + 2 + 2 = 6 ones.

Step 4: 40 + 6 = 46.

Answer: 46 — 4 Tens, 6 Ones

(4) 7 ten-strips and 4 ones

Step 1: Count the ten-strips: there are 7 strips of 10.

Step 2: 7 tens = 70.

Step 3: Count the single squares: there are 4 ones.

Step 4: 70 + 4 = 74.

Answer: 74 — 7 Tens, 4 Ones

Learning Tasks 2 (page 25) — Count the objects and write the number

Count the stars

Step 1: Count the full rows: there are 5 rows of 10 stars.

Step 2: 5 rows of 10 = 50 stars.

Step 3: Count the scattered stars beside the rows: there are 10 stars.

Step 4: 50 + 10 = 60.

Answer: 60 stars

Learning Tasks 3 (page 25) — Write the missing numbers forwards

Row 1: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Row 2: 11, 12, 13, 14, 15, 16, 17, 18, 19, 20. Row 3: 21, 22, 23, 24, 25, 26, 27, 28, 29, 30. Row 4: 31, 32, 33, 34, 35, 36, 37, 38, 39, 40. Row 5: 41, 42, 43, 44, 45, 46, 47, 48, 49, 50. Row 6: 51, 52, 53, 54, 55, 56, 57, 58, 59, 60. Row 7: 61, 62, 63, 64, 65, 66, 67, 68, 69, 70. Row 8: 71, 72, 73, 74, 75, 76, 77, 78, 79, 80. Row 9: 81, 82, 83, 84, 85, 86, 87, 88, 89, 90. Row 10: 91, 92, 93, 94, 95, 96, 97, 98, 99, 100.

Learning Tasks 4 (page 25) — Write the missing numbers backwards

Row 1: 50, 49, 48, 47, 46, 45, 44, 43, 42, 41. Row 2: 40, 39, 38, 37, 36, 35, 34, 33, 32, 31. Row 3: 30, 29, 28, 27, 26, 25, 24, 23, 22, 21. Row 4: 20, 19, 18, 17, 16, 15, 14, 13, 12, 11. Row 5: 10, 9, 8, 7, 6, 5, 4, 3, 2, 1.

Ordering Numbers (page 26)

Arranging the numbers from smaller to bigger is increasing order.

Example: 3, 4, 5, 6, 7, 8. Arranging the numbers from bigger to smaller is decreasing order. Example: 46, 45, 44, 43, 42, 41. Key point: Increasing order of numbers is also called ascending order. Decreasing order of numbers is also called descending order.

Learning Task 1 (page 27) — Arrange the numbers in increasing order

Q: (a) 25, 22, 24, 17 in increasing order

A: 17, 22, 24, 25 — start from the smallest (17), then 22, then 24, then the biggest (25).

Q: (b) 50, 40, 70, 60 in increasing order

A: 40, 50, 60, 70 — start from the smallest (40), then 50, then 60, then the biggest (70).

Q: (c) 28, 27, 29, 26 in increasing order

A: 26, 27, 28, 29 — start from the smallest (26), then 27, then 28, then the biggest (29).

Learning Task 2 (page 27) — Arrange the numbers in decreasing order

Q: (a) 72, 73, 74, 71 in decreasing order

A: 74, 73, 72, 71 — start from the biggest (74), then 73, then 72, then the smallest (71).

Q: (b) 75, 80, 70, 65 in decreasing order

A: 80, 75, 70, 65 — start from the biggest (80), then 75, then 70, then the smallest (65).

Q: (c) 91, 92, 89, 90 in decreasing order

A: 92, 91, 90, 89 — start from the biggest (92), then 91, then 90, then the

smallest (89).

Before, After and Between (page 28)

Look at the number of beads: 4, 3, 5. Arrange the number of beads in increasing order: 3, 4, 5. 3 Comes before 4. 4 Comes between 3 and 5. 5 Comes after 4.

Learning Task 1 (page 28) — Write the number that comes before the given number

Q: Before 25

A: 24 (given as the worked example in the book).

Q: Before 46

A: 45 — one less than 46 is 45.

Q: Before 68

A: 67 — one less than 68 is 67.

Q: Before 54

A: 53 — one less than 54 is 53.

Learning Task 2 (page 29) — Write the number that comes after the given number

Q: After 30

A: 31 (given as the worked example in the book).

Q: After 53

A: 54 — one more than 53 is 54.

Q: After 72

A: 73 — one more than 72 is 73.

Q: After 12

A: 13 — one more than 12 is 13.

Learning Task 3 (page 29) — Write the number between the given numbers

Q: (a) 7 ___ 9

A: 8 — 8 comes between 7 and 9.

Q: (b) 18 ___ 20

A: 19 — 19 comes between 18 and 20.

Q: (c) 26 ___ 28

A: 27 — 27 comes between 26 and 28.

Q: (d) 47 ___ 49

A: 48 — 48 comes between 47 and 49.

Q: (e) 65 ___ 67

A: 66 — 66 comes between 65 and 67.

Q: (f) 89 ___ 91

A: 90 — 90 comes between 89 and 91.

Q: (g) 95 ___ 97

A: 96 — 96 comes between 95 and 97.

Q: (h) 57 ___ 59

A: 58 — 58 comes between 57 and 59.

More, Less and Equal (page 30)

Bano has 4 candies. Amjad has 6 candies. Amjad has more candies. Bano has less candies. Then Amjad has 4 candies and Bano has 4 candies. So Amjad and Bano have equal candies.

Learning Task 1 (page 30) — Tick (correct) more objects and

cross (wrong) less objects

Q: (a) Left box 5 fish, right box 6 fish

A: Left box (5 fish) gets a cross — it has less. Right box (6 fish) gets a tick — it has more.

Ordinal Numbers 1st to 10th (page 31)

Ordinal number describe the order or position of things. Let's observe the running race and describe their position order like 'first', 'second', 'third' up to tenth. The race positions from front to back are: First (1st Position), Second (2nd Position), Third (3rd Position), Fourth (4th Position), Fifth (5th Position), Sixth (6th Position), Seventh (7th Position), Eighth (8th Position), Ninth (9th Position), Tenth (10th Position).

Learning Task 1 (page 32) — Tick the picture to show the given position

Q: (a) 3rd leaf

A: Tick the 3rd leaf from the left (given as the worked example in the book).

Q: (b) 5th flower

A: Tick the 5th flower from the left — the last flower in the row.

Q: (c) 4th cup

A: Tick the 4th cup from the left.

Q: (d) 7th butterfly

A: Tick the 7th butterfly from the left — the last butterfly in the row.

Learning Task 2 (page 32) — Colour the pictures to show the given ordinal number from left to right

Q: (a) 8th book

A: Colour the 8th book from the left — the last book in the row. (Note: the book already shows the 6th book coloured red; that coloured book is not the 8th — the 8th book is the last one.)

Q: (b) 5th ice cream

A: Colour the 5th ice cream from the left.

Q: (c) 6th fish

A: Colour the 6th fish from the left — the last fish in the row.

Learning Task 3 (page 33) — Write the position of coloured object

Q: (a) Red flower

A: 5th — the red flower is the 5th flower from the left (labelled in the book).

Q: (b) Red candle

A: 9th — counting from the left, the red candle is the 9th candle in the row of 10.

Q: (c) Coloured ball

A: 3rd — the coloured ball is the 3rd ball from the left.

I have Learnt (page 33): Reading and writing number up to 99 in numerals and words. Counting objects up to 99 forward and backwards. Identifying the numbers before, after and between the numbers. Ordering the numbers in increasing and decreasing order. Identifying the place value of specific digit in 2-digit numbers. Identifying the position of objects using ordinal number (First, Second, Third…). Identify more or less by comparing the number of objects in two groups. Vocabulary used: Numbers, more, less, counting, forward, backward, place value, ones,

tens, increasing, decreasing, ordinal number, ascending, descending.

Review Exercise 1 (page 34) — Choose the correct answer; I)

Count and tick the correct number

How many stars? (options: 6, 7, 8)

- 6
- 7
- ✓ 8

How many balls? (options: 5, 6, 7)

- 5
- ✓ 6
- 7

How many ducks? (options: 9, 10, 11)

- 9
- ✓ 10
- 11

Review Exercise 2 (page 34) — Write the missing numbers

Row 1: 9, 10, 11, 12, 13, 14, 15, 16, 17, 18. Row 2: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36. Row 3: 54, 55, 56, 57, 58, 59, 60, 61, 62, 63.

Review Exercise 3 (page 34) — Match the columns

Number — word

17 — Seventeen

20 — Twenty

25 — Twenty five

31 — Thirty one

40 — Forty

57 — Fifty seven

Review Exercise 4 (page 35) — Write the following numbers in words

Number — word

13 — Thirteen

19 — Nineteen

25 — Twenty five

30 — Thirty

38 — Thirty eight

45 — Forty five

59 — Fifty nine

62 — Sixty two

86 — Eighty six

99 — Ninety nine

Review Exercise 5 (page 35) — Arrange the numbers in increasing and decreasing order

Q: (i) 17, 5, 30, 22 — Increasing order

A: 5, 17, 22, 30.

Q: (i) 17, 5, 30, 22 — Decreasing order

A: 30, 22, 17, 5.

Q: (ii) 78, 26, 39, 50 — Increasing order

A: 26, 39, 50, 78.

Q: (ii) 78, 26, 39, 50 — Decreasing order

A: 78, 50, 39, 26.

Review Exercise 6 (page 36) — Write the place value of the given balls

Tens box: 2 rows of 10 balls; Ones box: 5 balls

Step 1: Count the balls in the Tens box: there are 2 rows of 10 balls.

Step 2: 2 rows of 10 = 20, so there are 2 tens.

Step 3: Count the balls in the Ones box: there are 5 single balls.

Step 4: 5 single balls = 5 ones.

Step 5: 2 tens + 5 ones = 25.

Answer: 25 — the place value of 2 is tens and the place value of 5 is ones

Review Exercise 7 (page 36) — Colour the 3rd, 5th, and 6th ordinal number form (left to right)

Q: 6 cups in a row — which cups to colour?

A: Colour the 3rd cup, the 5th cup and the 6th cup from the left.

Unit 2 — Numbers Operations (pages 37 to 63)

Students' learning outcomes (page 37): After completing this unit, students will be able to: Find recall and use of addition and subtraction facts to 20; Compare numbers to find how many more and how many less; Add and subtract numbers mentally and in written form including: Up to three 1-digit numbers, 2-digit numbers and tens; Add and subtract

2-digit numbers mentally and in written form including: 2 digit numbers and 1-digit numbers (without regrouping), Two 2-digit numbers (without regrouping); Solve real world problem with additions and subtraction using concrete objects and pictorial representations (involving missing numbers, money, quantities and measures).

How Many More (page 38)

Let's compare number of pencils and sharpeners. Numbers of pencils = 5. Number of sharpeners = 3. How many pencils are more than sharpeners? = 2. 5 is more than 3. 5 is 2 more than 3.

Activity (page 38): Let us compare 5 and 8. Count the numbers from 5 to 8. 8 is 3 steps forward from 5. So 8 is 3 more than 5.

Learning Task 1 (page 39) — Compare the following group of objects and tell how many more one is from other

(a) 9 pencils and 6 sharpeners

Step 1: Count the pencils: there are 9.

Step 2: Count the sharpeners: there are 6.

Step 3: 9 - 6 = 3.

Answer: 9 is 3 more than 6 (given as the worked example in the book)

(b) 13 ice creams and 8 ice creams

Step 1: Count the first group: there are 13 ice creams.

Step 2: Count the second group: there are 8 ice creams.

Step 3: 13 - 8 = 5.

Answer: 13 is 5 more than 8

(c) 15 footballs and 7 balloons

Step 1: Count the footballs: there are 15.

Step 2: Count the balloons: there are 7.

Step 3: 15 - 7 = 8.

Answer: 15 is 8 more than 7

(d) 14 books and 11 dolls

Step 1: Count the books: there are 14.

Step 2: Count the dolls: there are 11.

Step 3: 14 - 11 = 3.

Answer: 14 is 3 more than 11

How Many Less (page 40)

Let's compare number of flowers and butterflies. Numbers of flowers = 5. Number of butterflies = 2. How many butterflies are less than flowers? = 3. 2 is less than 5. 2 is 3 less than 5.

Let's compare number of pencils and sharpeners. Numbers of pencils = 5. Number of sharpeners = 4. How many pencils are more than sharpeners? The book's box prints 2 here, but 5 - 4 = 1, so the correct value is 1: 5 is 1 more than 4. 4 is less than 5. 4 is 1 less than 5. (The book's own text below the box confirms 1 — the box value 2 is a book error, flagged here.)

Learning Task 1 (page 41) — Compare the group of objects and tell how many less one is from other

(a) 7 bats and 4 balls

Step 1: Count the bats: there are 7.

Step 2: Count the balls: there are 4.

Step 3: 7 - 4 = 3.

Answer: 4 is 3 less than 7 (the book's printed frame reads '7 is ___ less than 4', which is reversed — the smaller number comes first in a 'less than' sentence)

(b) 5 strawberries and 3 apples

Step 1: Count the strawberries: there are 5.

Step 2: Count the apples: there are 3.

Step 3: 5 - 3 = 2.

Answer: 3 is 2 less than 5 (the book's printed frame reads '5 is ___ less than 3', which is reversed)

(c) 14 books and 11 dolls

Step 1: Count the books: there are 14.

Step 2: Count the dolls: there are 11.

Step 3: 14 - 11 = 3.

Answer: 11 is 3 less than 14

Addition of 1-Digit Number (page 42)

Addition means to put things together. 5 candies and 4 candies makes 9 candies: 5 + 4 = 9. 3 dates, 5 dates and 4 dates makes 12 dates. We can write it as: 3 + 5 + 4 = 12. Key point: The symbol '+' is used for addition and '=' is used for equality.

Learning Task 1 (page 43) — Count, write numbers and add

(a) 5 flowers + 3 flowers

Step 1: Count the first group: 5 flowers.

Step 2: Count the second group: 3 flowers.

Step 3: 5 + 3 = 8.

Answer: 8 (given as the worked example in the book)

(b) 4 books + 2 books + 3 books

Step 1: Count the groups: 4 books, 2 books and 3 books.

Step 2: 4 + 2 = 6.

Step 3: 6 + 3 = 9.

Answer: 9 books

(c) 4 dolls + 3 dolls

Step 1: Count the first group: 4 dolls.

Step 2: Count the second group: 3 dolls.

Step 3: 4 + 3 = 7.

Answer: 7 dolls

Learning Task 2 (page 43) — Add the following

(a) 7 + 2

Step 1: Start at 7.

Step 2: Count 2 steps forward: 8, 9.

Answer: 9

(b) 9 + 0

Step 1: Adding zero leaves the number unchanged.

Step 2: 9 + 0 = 9.

Answer: 9

(c) 6 + 3

Step 1: Start at 6.

Step 2: Count 3 steps forward: 7, 8, 9.

Answer: 9

Addition of 2-Digit Number and Tens (page 44)

Adding 12 and 20

Step 1: Write the numbers in Tens and Ones: 12 = 1 ten and 2 ones; 20 = 2 tens and 0 ones.

Step 2: Add the tens: 1 ten + 2 tens = 3 tens.

Step 3: Add the ones: 2 ones + 0 ones = 2 ones.

Step 4: 3 tens and 2 ones = 32.

Answer: 32

Key point (page 44): When zero '0' is added to any number, the number remains same.

Adding 27 and 50

Step 1: Write the numbers in Tens and Ones: 27 = 2 tens and 7 ones; 50 = 5 tens and 0 ones.

Step 2: Add the tens: 2 tens + 5 tens = 7 tens.

Step 3: Add the ones: 7 ones + 0 ones = 7 ones.

Step 4: 7 tens and 7 ones = 77.

Answer: 77

Addition of 2-Digit Number (page 45) — Adding 32 and 13

32 + 13

Step 1: Write the numbers in Tens and Ones: 32 = 3 tens and 2 ones; 13 = 1 ten and 3 ones.

Step 2: Add the tens: 3 tens + 1 ten = 4 tens.

Step 3: Add the ones: 2 ones + 3 ones = 5 ones.

Step 4: 4 tens and 5 ones = 45.

Answer: 45

Learning Task 1 (page 46) — Add

35 + 10

Step 1: Tens: 3 + 1 = 4 tens.

Step 2: Ones: 5 + 0 = 5 ones.

Answer: 45

28 + 30

Step 1: Tens: 2 + 3 = 5 tens.

Step 2: Ones: 8 + 0 = 8 ones.

Answer: 58

73 + 20

Step 1: Tens: 7 + 2 = 9 tens.

Step 2: Ones: 3 + 0 = 3 ones.

Answer: 93

50 + 40

Step 1: Tens: 5 + 4 = 9 tens.

Step 2: Ones: 0 + 0 = 0 ones.

Answer: 90

48 + 20

Step 1: Tens: 4 + 2 = 6 tens.

Step 2: Ones: 8 + 0 = 8 ones.

Answer: 68

49 + 50

Step 1: Tens: 4 + 5 = 9 tens.

Step 2: Ones: 9 + 0 = 9 ones.

Answer: 99

37 + 60

Step 1: Tens: 3 + 6 = 9 tens.

Step 2: Ones: 7 + 0 = 7 ones.

Answer: 97

17 + 80

Step 1: Tens: 1 + 8 = 9 tens.

Step 2: Ones: 7 + 0 = 7 ones.

Answer: 97

Mental Addition (page 47)

Sidra has 34 colour pencils, her brother gives her 12 more. How many pencils she has altogether? Let's add the number 34 and 12 mentally: Tens 3, Ones 4; plus Tens 1, Ones 2; gives Tens 4, Ones 6. Sidra has 46 pencils.

Learning Task 1 (page 47) — There are 45 flowers in a vase. Add 13 more flowers. How many flowers are there in vase now? Solve mentally.

45 + 13

Step 1: Tens: 4 + 1 = 5 tens.

Step 2: Ones: 5 + 3 = 8 ones.

Step 3: 5 tens and 8 ones = 58.

Answer: 58 flowers

Learning Task 2 (page 47) — Add the number mentally

(a) 25 + 23

Step 1: Tens: 2 + 2 = 4 tens.

Step 2: Ones: 5 + 3 = 8 ones.

Answer: 48

(b) 18 + 51

Step 1: Tens: 1 + 5 = 6 tens.

Step 2: Ones: 8 + 1 = 9 ones.

Answer: 69

(c) 30 + 19

Step 1: Tens: 3 + 1 = 4 tens.

Step 2: Ones: 0 + 9 = 9 ones.

Answer: 49

(d) 64 + 13

Step 1: Tens: 6 + 1 = 7 tens.

Step 2: Ones: 4 + 3 = 7 ones.

Answer: 77

(e) 81 + 15

Step 1: Tens: 8 + 1 = 9 tens.

Step 2: Ones: 1 + 5 = 6 ones.

Answer: 96

(f) 63 + 35

Step 1: Tens: 6 + 3 = 9 tens.

Step 2: Ones: 3 + 5 = 8 ones.

Answer: 98

(g) 41 + 37

Step 1: Tens: 4 + 3 = 7 tens.

Step 2: Ones: 1 + 7 = 8 ones.

Answer: 78

(h) 56 + 32

Step 1: Tens: 5 + 3 = 8 tens.

Step 2: Ones: 6 + 2 = 8 ones.

Answer: 88

Real Life Problems (page 48)

Husnain has 4 balls. His sister gave him 2 more balls. How many balls he has?

Step 1: Husnain has balls = 4.

Step 2: Sister gave him more balls = 2.

Step 3: 4 + 2 = 6.

Answer: 6 balls

Areeba has 7 sharpeners. Nimra has 6 sharpeners. How many sharpeners do they have altogether?

Step 1: Areeba's sharpeners = 7.

Step 2: Nimra's sharpeners = 6.

Step 3: 7 + 6 = 13.

Answer: 13 sharpeners

Yasir has Rs. 10. He gets Rs. 10 more. How many rupees does he has altogether?

Step 1: Yasir had = Rs. 10.

Step 2: He was given more = Rs. 10.

Step 3: 10 + 10 = 20.

Answer: Rs. 20

Learning Task 1 (page 49) — Kamran has 5 footballs and 9 cricket balls. How many balls does he has altogether?

5 + 10 (book's working box and pictures)

Step 1: The question text says 5 footballs and 9 cricket balls, but the book's own working box shows 5 and 10, and the pictures show 5 soccer balls and 10 red balls.

Step 2: Following the book's working box and pictures: 5 + 10 = 15.

Step 3: Note: if the question text (9 cricket balls) is followed instead, the total would be 5 + 9 = 14. The mismatch is in the book itself and is flagged here.

Answer: 15 balls (from the book's box and pictures; 14 from the question text)

Learning Task 2 (page 49) — There are 12 eggs in the basket. Mother puts 10 more in it. How many eggs are there altogether?

12 + 10

Step 1: Eggs in the basket = 12 (1 ten and 2 ones).

Step 2: More eggs = 10 (1 ten and 0 ones).

Step 3: Tens: 1 + 1 = 2 tens. Ones: 2 + 0 = 2 ones.

Step 4: 2 tens and 2 ones = 22.

Answer: 22 eggs

Learning Task 3 (page 49) — Nasreen has 10 red beads and 15 blue beads. How many beads does she have in total?

10 + 15

Step 1: Red beads = 10 (1 ten and 0 ones).

Step 2: Blue beads = 15 (1 ten and 5 ones).

Step 3: Tens: 1 + 1 = 2 tens. Ones: 0 + 5 = 5 ones.

Step 4: 2 tens and 5 ones = 25.

Answer: 25 beads

Subtraction (page 50)

Subtraction means to take away from the collection. 5 Biscuits, take away 2: Biscuits left = 3 (5 - 2 = 3). 7 Rose flowers, take away 5: Rose flowers left = 2 (7 - 5 = 2). When we take away we subtract.

Subtraction (page 51)

Apples on the tree = 7. Apples fall from the tree = 2. How many apples left = 5. This can be written as: 7 - 2 = 5. Key point: '-' is subtraction symbol. '-' is also called minus. Total candies are 4. Crossed candies are 2. Remaining candies are 2 (4 - 2 = 2).

Subtraction of 1-Digit Numbers (page 52)

Look at these mangoes: Total mangoes are = 7. Take away mangoes = 4.

Remaining mangoes = 3. We can write it as: 7 - 4 = 3.

Subtract (page 52)

(a) 9 - 5

Step 1: Start at 9.

Step 2: Count 5 steps back: 8, 7, 6, 5, 4.

Answer: 4 (given as the worked example in the book)

(b) 8 - 3

Step 1: Start at 8.

Step 2: Count 3 steps back: 7, 6, 5.

Answer: 5

(c) 7 - 3

Step 1: Start at 7.

Step 2: Count 3 steps back: 6, 5, 4.

Answer: 4

(d) 5 - 4

Step 1: Start at 5.

Step 2: Count 4 steps back: 4, 3, 2, 1.

Answer: 1

(e) 5 - 3

Step 1: Start at 5.

Step 2: Count 3 steps back: 4, 3, 2.

Answer: 2

(f) 4 - 2

Step 1: Start at 4.

Step 2: Count 2 steps back: 3, 2.

Answer: 2

(g) 7 - 2

Step 1: Start at 7.

Step 2: Count 2 steps back: 6, 5.

Answer: 5

(h) 8 - 5

Step 1: Start at 8.

Step 2: Count 5 steps back: 7, 6, 5, 4, 3.

Answer: 3

Subtract 1-Digit Numbers from 2-Digit Numbers (page 53)

Subtract 4 from 17

Step 1: Write the numbers vertically.

Step 2: Subtract ones from ones: 7 ones - 4 ones = 3 ones.

Step 3: Subtract tens from tens: 1 ten - 0 ten = 1 ten.

Step 4: 1 ten and 3 ones = 13.

Answer: 13

Subtract 8 from 28

Step 1: Write the numbers vertically.

Step 2: Subtract ones from ones: 8 ones - 8 ones = 0 ones.

Step 3: Subtract tens from tens: 2 tens - 0 ten = 2 tens.

Step 4: 2 tens and 0 ones = 20.

Answer: 20

Key point (page 53): When same number are subtracted, the answer is

always Zero (0).

Learning Task (page 54) — Subtract

27 - 5

Step 1: Ones: 7 - 5 = 2 ones.

Step 2: Tens: 2 - 0 = 2 tens.

Answer: 22

38 - 7

Step 1: Ones: 8 - 7 = 1 one.

Step 2: Tens: 3 - 0 = 3 tens.

Answer: 31

49 - 6

Step 1: Ones: 9 - 6 = 3 ones.

Step 2: Tens: 4 - 0 = 4 tens.

Answer: 43

18 - 4

Step 1: Ones: 8 - 4 = 4 ones.

Step 2: Tens: 1 - 0 = 1 ten.

Answer: 14

55 - 4

Step 1: Ones: 5 - 4 = 1 one.

Step 2: Tens: 5 - 0 = 5 tens.

Answer: 51

69 - 8

Step 1: Ones: 9 - 8 = 1 one.

Step 2: Tens: 6 - 0 = 6 tens.

Answer: 61

76 - 3

Step 1: Ones: 6 - 3 = 3 ones.

Step 2: Tens: 7 - 0 = 7 tens.

Answer: 73

16 - 4

Step 1: Ones: 6 - 4 = 2 ones.

Step 2: Tens: 1 - 0 = 1 ten.

Answer: 12

88 - 7

Step 1: Ones: 8 - 7 = 1 one.

Step 2: Tens: 8 - 0 = 8 tens.

Answer: 81

Subtraction of Tens from 2-Digit Numbers (page 55)

Subtract 10 from 26

Step 1: Write the numbers vertically: 26 = 2 tens and 6 ones; 10 = 1 ten and 0 ones.

Step 2: Subtract tens: 2 tens - 1 ten = 1 ten.

Step 3: Subtract ones: 6 ones - 0 ones = 6 ones.

Step 4: 1 ten and 6 ones = 16.

Answer: 16

Subtract 20 from 45

Step 1: Write the numbers vertically: 45 = 4 tens and 5 ones; 20 = 2 tens

and 0 ones.

Step 2: Subtract tens: 4 tens - 2 tens = 2 tens.

Step 3: Subtract ones: 5 ones - 0 ones = 5 ones.

Step 4: 2 tens and 5 ones = 25.

Answer: 25

Key fact (page 55): When Zero (0) is subtracted from any other number, the answer is number itself.

Subtraction of 2-Digit Numbers from 2-Digit Numbers (page 56)

Subtract 32 from 57

Step 1: Write the numbers vertically: 57 = 5 tens and 7 ones; 32 = 3 tens and 2 ones.

Step 2: Subtract tens: 5 tens - 3 tens = 2 tens.

Step 3: Subtract ones: 7 ones - 2 ones = 5 ones.

Step 4: 2 tens and 5 ones = 25.

Answer: 25

Subtract 23 from 46

Step 1: Write the numbers vertically: 46 = 4 tens and 6 ones; 23 = 2 tens and 3 ones.

Step 2: Subtract tens: 4 tens - 2 tens = 2 tens.

Step 3: Subtract ones: 6 ones - 3 ones = 3 ones.

Step 4: 2 tens and 3 ones = 23.

Answer: 23

Learning Task (page 57) — Subtract

(1) 17 - 10

Step 1: Tens: 1 - 1 = 0 tens.

Step 2: Ones: 7 - 0 = 7 ones.

Answer: 7

(2) 11 - 10

Step 1: Tens: 1 - 1 = 0 tens.

Step 2: Ones: 1 - 0 = 1 one.

Answer: 1

(3) 37 - 20

Step 1: Tens: 3 - 2 = 1 ten.

Step 2: Ones: 7 - 0 = 7 ones.

Answer: 17

(4) 65 - 30

Step 1: Tens: 6 - 3 = 3 tens.

Step 2: Ones: 5 - 0 = 5 ones.

Answer: 35

(5) 29 - 27

Step 1: Tens: 2 - 2 = 0 tens.

Step 2: Ones: 9 - 7 = 2 ones.

Answer: 2

(6) 26 - 14

Step 1: Tens: 2 - 1 = 1 ten.

Step 2: Ones: 6 - 4 = 2 ones.

Answer: 12

(7) 55 - 42

Step 1: Tens: 5 - 4 = 1 ten.

Step 2: Ones: 5 - 2 = 3 ones.

Answer: 13

(8) 94 - 53

Step 1: Tens: 9 - 5 = 4 tens.

Step 2: Ones: 4 - 3 = 1 one.

Answer: 41

(9) 38 - 18

Step 1: Tens: 3 - 1 = 2 tens.

Step 2: Ones: 8 - 8 = 0 ones.

Answer: 20

Mental Subtraction (page 58)

Haris has Rs. 36. He gave Rs. 15 to his sister. How many rupees are left with him? Let's subtract numbers mentally. Step I: Separate tens and ones: 36 is 30 + 6, 15 is 10 + 5. Step II: Subtract tens and ones separately: 30 - 10 = 20, 6 - 5 = 1. Step III: Add the result: 20 + 1 = 21. So Rs. 21/- are left.

Learning Task 1 (page 58) — Sidra has 29 pencils. She gave 14 to her friend. How many pencils are left? Solve mentally.

29 - 14

Step 1: Step I: Separate tens and ones: 29 is 20 + 9, 14 is 10 + 4.

Step 2: Step II: Subtract tens and ones separately: 20 - 10 = 10, 9 - 4 = 5.

Step 3: Step III: Add the result: 10 + 5 = 15.

Answer: 15 pencils

Learning Task 2 (page 58) — Subtract mentally

(a) 14 - 6

Step 1: Ones: 14 - 6 = 8.

Answer: 8

(b) 48 - 37

Step 1: Tens: 4 - 3 = 1 ten.

Step 2: Ones: 8 - 7 = 1 one.

Answer: 11

(c) 47 - 9

Step 1: Ones: 47 - 9: 47 - 7 = 40, then 40 - 2 = 38.

Answer: 38

(d) 86 - 55

Step 1: Tens: 8 - 5 = 3 tens.

Step 2: Ones: 6 - 5 = 1 one.

Answer: 31

(e) 58 - 8

Step 1: Ones: 8 - 8 = 0.

Step 2: Tens: 5 - 0 = 5 tens.

Answer: 50

(f) 76 - 35

Step 1: Tens: 7 - 3 = 4 tens.

Step 2: Ones: 6 - 5 = 1 one.

Answer: 41

Real Life Problems (page 59)

There are 8 kids in a family. 5 of them are boys. How many girls are there in the family?

Step 1: Total kids = 8.

Step 2: Boys = 5.

Step 3: 8 - 5 = 3.

Answer: There are 3 girls in the family

Learning Task 1 (page 59) — Salman had 35 balloons. 12 got burst. How many balloons are left?

35 - 12

Step 1: Total balloons = 35 (3 tens and 5 ones).

Step 2: Balloons burst = 12 (1 ten and 2 ones).

Step 3: Tens: 3 - 1 = 2 tens. Ones: 5 - 2 = 3 ones.

Step 4: 2 tens and 3 ones = 23.

Answer: 23 balloons left

Learning Task 2 (page 59) — A storybook has 48 pages. Yasir reads 26 pages. How many pages are left to read?

48 - 26

Step 1: Total pages = 48 (4 tens and 8 ones).

Step 2: Yasir read = 26 (2 tens and 6 ones).

Step 3: Tens: 4 - 2 = 2 tens. Ones: 8 - 6 = 2 ones.

Step 4: 2 tens and 2 ones = 22.

Answer: 22 pages left to read

Learning Task 3 (page 60) — There were 40 Rupees in the wallet. 20 Rupees were used in buying a toy. How many rupees are left?

40 - 20

Step 1: Total rupees = 40 (4 tens and 0 ones).

Step 2: Toy bought = 20 (2 tens and 0 ones).

Step 3: Tens: 4 - 2 = 2 tens. Ones: 0 - 0 = 0 ones.

Answer: Rs. 20 left

Learning Task 4 (page 60) — Denar had 18 birds. 7 birds flew away. How many birds are left?

18 - 7

Step 1: Total birds = 18.

Step 2: Flew = 7.

Step 3: 18 - 7 = 11.

Answer: 11 birds left

I have Learnt (page 61): Comparing numbers up to 20. Identifying the symbols of addition (+) and subtraction (-). Adding and subtracting of 1-digit numbers. Adding and subtracting of 2-digit numbers. Subtracting tens from 2-digit numbers. Adding and Subtracting numbers (up to 20) using mental strategies. Solving real world problems of addition and Subtraction. Vocabulary used: Addition, sum, equal, Total, Subtraction, left, Difference, take away, more less.

Review Exercise 1 (page 61) — Choose the correct answer

The symbol '+' is used for (options: Equality, Subtraction, Addition, Comparing)

Equality

Subtraction

✓ Addition

Comparing

Review 1 (ii): 12 + 8 = 20. Note: the book's options are 128, 92, 22 and 28 — none of them is 20, so the correct answer is not among the printed options. This is a book error, flagged here; the correct sum is 20.

4 + ___ = 9 (options: 4, 5, 9, 13)

4

✓ 5

9

13

8 is ___ less than 13 (options: 2, 3, 5, 9)

2

3

✓ 5

9

12 - 10 = ___ (options: 2, 08, 10, 12)

✓ 2

08

10

12

Review Exercise 2 (page 62) — Compare the following group of object and tell how many more one is from other

7 apples and 5 bananas

Step 1: Count the apples: 7.

Step 2: Count the bananas: 5.

Step 3: 7 - 5 = 2.

Answer: 7 is 2 more than 5

11 glasses and 7 cups

Step 1: Count the glasses: 11.

Step 2: Count the cups: 7.

Step 3: 11 - 7 = 4.

Answer: 11 is 4 more than 7

4 dates and 8 candies

Step 1: Count the dates: 4.

Step 2: Count the candies: 8.

Step 3: 8 - 4 = 4.

Answer: 4 is 4 less than 8

12 arrows and 15 stars

Step 1: Count the arrows: 12.

Step 2: Count the stars: 15.

Step 3: 15 - 12 = 3.

Answer: 12 is 3 less than 15

Review Exercise 3 (page 63) — Count and add the numbers

(i) 6 stars + 3 stars

Step 1: Count the first box: 6 stars.

Step 2: Count the second box: 3 stars.

Step 3: 6 + 3 = 9.

Answer: 9

(ii) 5 flowers + 3 flowers

Step 1: Count the first box: 5 flowers.

Step 2: Count the second box: 3 flowers.

Step 3: 5 + 3 = 8.

Answer: 8

Review Exercise 4 (page 63) — Add the following

53 + 43

Step 1: Tens: 5 + 4 = 9 tens.

Step 2: Ones: 3 + 3 = 6 ones.

Answer: 96

15 + 64

Step 1: Tens: 1 + 6 = 7 tens.

Step 2: Ones: 5 + 4 = 9 ones.

Answer: 79

83 + 12

Step 1: Tens: 8 + 1 = 9 tens.

Step 2: Ones: 3 + 2 = 5 ones.

Answer: 95

23 + 64

Step 1: Tens: 2 + 6 = 8 tens.

Step 2: Ones: 3 + 4 = 7 ones.

Answer: 87

Review Exercise 5 (page 63) — Subtract the following

26 - 13

Step 1: Tens: 2 - 1 = 1 ten.

Step 2: Ones: 6 - 3 = 3 ones.

Answer: 13

57 - 46

Step 1: Tens: 5 - 4 = 1 ten.

Step 2: Ones: 7 - 6 = 1 one.

Answer: 11

48 - 33

Step 1: Tens: 4 - 3 = 1 ten.

Step 2: Ones: 8 - 3 = 5 ones.

Answer: 15

86 - 50

Step 1: Tens: 8 - 5 = 3 tens.

Step 2: Ones: 6 - 0 = 6 ones.

Answer: 36

Review Exercise 6 (page 63) — Abbas had 12 Toy cars, He bought 10 more. How many cars does he has altogether?

12 + 10

Step 1: Abbas had Toys = 12.

Step 2: Bought more = 10.

Step 3: 12 + 10 = 22.

Answer: 22 cars altogether

Review Exercise 7 (page 63) — Ahmad had 25 apples. He gave 13 to his sister. How many apples are left?

25 - 13

Step 1: Ahmad had 25 apples.

Step 2: Gave to sister = 13.

Step 3: 25 - 13 = 12.

Answer: 12 apples left

Unit 3 — Multiplication and Division (pages 64 to 83)

Students' learning outcomes (page 64): After completing this unit, students will be able to: count and write in 2's, 5's and 10's using concrete objects and pictorial representations such as the number line and hundred square grid; recognize counting in 2's, 5's and 10's as the multiplication tables of two, five and ten; recognize multiplication as repeated addition using concrete objects and pictorial representations (groups and arrays); recognize that the multiplication of any two numbers can be done in any order; recognize division using groups, arrays and sharing; solve simple real-life problems involving multiplication and division.

Count in 2's, 5's and 10's (pages 65-67) — Skip counting

Count in 2's — number in words

2 — Two

4 — Four

6 — Six

8 — Eight

10 — Ten

12 — Twelve

14 — Fourteen

16 — Sixteen

18 — Eighteen

20 — Twenty

Count in 5's — number in words

5 — Five

10 — Ten

15 — Fifteen

20 — Twenty

25 — Twenty Five

30 — Thirty

35 — Thirty Five

40 — Forty

45 — Forty Five

50 — Fifty

Count in 10's — number in words

10 — Ten

20 — Twenty

30 — Thirty

40 — Forty

50 — Fifty

60 — Sixty

70 — Seventy

80 — Eighty

90 — Ninety

100 — One Hundred

Key point (page 65): Skip counting is counting in 2's, 5's and 10's.

Multiplication (page 68) — Repeated addition

There are 4 groups of apples and each group has 2 apples. 2 + 2 + 2 + 2 = 8. We can write it as 4 x 2 = 8. The sign 'x' means multiplication. It is repeated addition. We read 4 x 2 = 8 as '4 multiply 2 is equal to 8' or '4 times 2 is 8'.

3 + 3

Step 1: There are 2 groups of 3.

Step 2: 3 + 3 = 6.

Answer: 2 x 3 = 6

5 + 5

Step 1: There are 2 groups of 5.

Step 2: 5 + 5 = 10.

Answer: 2 x 5 = 10

Learning Task (page 69) — How many are there altogether?

1. Mangoes: 3 groups of 2

Step 1: Count the first group: 2 mangoes.

Step 2: Count the second group: 2 mangoes.

Step 3: Count the third group: 2 mangoes.

Step 4: 2 + 2 + 2 = 6.

Step 5: 3 times 2 = 6.

Answer: 3 x 2 = 6 mangoes altogether

2. Toy cars: groups of 4, 4 and 8

Step 1: Count the first group: 4 cars.

Step 2: Count the second group: 4 cars.

Step 3: Count the third group: 8 cars.

Step 4: 4 + 4 + 8 = 16.

Step 5: There are 16 cars altogether. NOTE (book error, flagged): the book's frame under this picture prints '4 + 4', but the picture actually shows 16 cars (4 + 4 + 8); the counted total 16 is used here.

Answer: 16 toy cars altogether

Order in Multiplication (page 70) — The order does not change the product

First array: 5 rows and 3 columns: 5 x 3 = 15. Second array: 3 rows and 5 columns: 3 x 5 = 15. So 5 x 3 = 3 x 5 = 15. Also 2 x 5 = 10 and 5 x 2 = 10, so 2 x 5 = 5 x 2. The order in which we multiply does not change the product.

Learning Task 1 (page 71) — Fill in the blanks

a) 2 x 1

Step 1: 2 times 1 is 2.

Answer: 2

b) 2 x 4

Step 1: 2 times 4 is 8.

Answer: 8

c) 5 + 5 + 5

Step 1: 5 + 5 = 10.

Step 2: 10 + 5 = 15.

Answer: 15

d) 5 x 6

Step 1: 5 times 6 is 30.

Answer: 30

e) 10 + 10

Step 1: 10 + 10 = 20.

Answer: 20

f) 10 x 4

Step 1: 10 times 4 is 40.

Answer: 40

Learning Task 2 (page 71) — Fill in the blanks (order in multiplication)

a) 2 x 7 = 7 x ___

Step 1: 2 x 7 = 14.

Step 2: 7 x 2 = 14.

Answer: 2

b) 2 x 5 = 5 x ___

Step 1: 2 x 5 = 10.

Step 2: 5 x 2 = 10.

Answer: 2

c) 5 x 3 = 3 x ___

Step 1: 5 x 3 = 15.

Step 2: 3 x 5 = 15.

Answer: 5

d) 10 x 7 = ___ x 10

Step 1: 10 x 7 = 70.

Step 2: 7 x 10 = 70.

Answer: 7

e) 10 x 4 = 4 x ___

Step 1: 10 x 4 = 40.

Step 2: 4 x 10 = 40.

Answer: 10

f) ___ x 6 = 6 x 5

Step 1: 6 x 5 = 30.

Step 2: 5 x 6 = 30.

Answer: 5

Learning Task 3 (page 71) — Complete the series

Series — completed

a) Count in 2's: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 — 2, 4, 6, 8, 10, 12, 14, 16, 18, 20

b) Count in 5's: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50 — 5, 10, 15, 20, 25, 30, 35, 40, 45, 50

c) Count in 10's: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 — 10, 20, 30, 40, 50, 60, 70, 80, 90, 100

Learning Task 4 (page 71) — Solve the following

3 x 2

Step 1: 3 times 2 is 6.

Answer: 6

4 x 2

Step 1: 4 times 2 is 8.

Answer: 8

7 x 2

Step 1: 7 times 2 is 14.

Answer: 14

4 x 5

Step 1: 4 times 5 is 20.

Answer: 20

3 x 5

Step 1: 3 times 5 is 15.

Answer: 15

6 x 5

Step 1: 6 times 5 is 30.

Answer: 30

5 x 10

Step 1: 5 times 10 is 50.

Answer: 50

8 x 10

Step 1: 8 times 10 is 80.

Answer: 80

Multiplication Table of 2 (page 72)

Table of 2

2 x 1 — 2

2 x 2 — 4

2 x 3 — 6

2 x 4 — 8

2 x 5 — 10

2 x 6 — 12

2 x 7 — 14

2 x 8 — 16

2 x 9 — 18

2 x 10 — 20

Multiplication Table of 5 (page 73)

Table of 5

5 x 1 — 5

5 x 2 — 10

5 x 3 — 15

5 x 4 — 20

5 x 5 — 25

5 x 6 — 30

5 x 7 — 35

5 x 8 — 40

5 x 9 — 45

5 x 10 — 50

Multiplication Table of 10 (page 74)

Table of 10

10 x 1 — 10

10 x 2 — 20

10 x 3 — 30

10 x 4 — 40

10 x 5 — 50

10 x 6 — 60

10 x 7 — 70

10 x 8 — 80

10 x 9 — 90

10 x 10 — 100

Activity 1 (page 75) — Fill the white spaces with the help of the table of 2

The wheel has the number 2 in the centre. Each white space is 2 times the inner number. Missing products: 2 x 2 = 4, 2 x 3 = 6, 2 x 4 = 8, 2 x 5 = 10, 2 x 7 = 14, 2 x 8 = 16, 2 x 10 = 20. (Given on the wheel: 2 x 1 = 2, 2 x 6 = 12, 2 x 9 = 18.)

Activity 2 (page 75) — Fill the white spaces with the help of the table of 5

The wheel has the number 5 in the centre. Each white space is 5 times the inner number. Missing products: 5 x 2 = 10, 5 x 3 = 15, 5 x 5 = 25, 5 x 7 = 35, 5 x 8 = 40, 5 x 10 = 50. (Given on the wheel: 5 x 1 = 5, 5 x 4 = 20, 5 x 6 = 30, 5 x 9 = 45.)

Activity 3 (page 76) — Fill the white spaces with the help of the table of 10

The wheel has the number 10 in the centre. Each white space is 10 times the inner number. Missing products: 10 x 2 = 20, 10 x 3 = 30, 10 x 4 = 40, 10 x 5 = 50, 10 x 7 = 70, 10 x 8 = 80, 10 x 10 = 100. (Given on the wheel: 10 x 1 = 10, 10 x 6 = 60, 10 x 9 = 90.)

Real Life Problems (page 76) — Computers on tables

4 tables, 2 computers on each table

Step 1: Number of tables = 4.

Step 2: Computers on each table = 2.

Step 3: 4 x 2 = 8.

Answer: 8 computers in the room

Real Life Problems (page 77)

2. 5 bags, 3 books in each bag

Step 1: Number of bags = 5.

Step 2: Books in each bag = 3.

Step 3: 5 x 3 = 15.

Answer: 15 books in the bags

Learning Task 1. 10 students in one class, 5 classes

Step 1: Students in one class = 10.

Step 2: Number of classes = 5.

Step 3: 10 x 5 = 50.

Answer: 50 students in 5 classes

Learning Task 2. One toy costs Rs. 10, price of 2 toys

Step 1: Price of one toy = Rs. 10.

Step 2: Number of toys = 2.

Step 3: 10 x 2 = 20.

Answer: Rs. 20 for 2 toys

Learning Task 3. 5 plates, 10 oranges on each plate

Step 1: Number of plates = 5.

Step 2: Oranges on one plate = 10.

Step 3: 5 x 10 = 50.

Answer: 50 oranges in total

Division as Repeated Subtraction (page 78)

We have 10 pencils and we share them into groups of 5. Keep taking away 5 until none are left: 10 - 5 = 5, then 5 - 5 = 0. We took away 5 two times, so 10 divided by 5 = 2. Order in division matters: 4 cakes shared among 2 children gives 4 divided by 2 = 2 cakes each, but 2 divided by 4 is not 2, so 4 divided by 2 is NOT equal to 2 divided by 4.

Real Life Problems (page 79) — Division

15 students divided into groups of 5

Step 1: Number of students = 15.

Step 2: Students in each group = 5.

Step 3: 15 divided by 5 = 3.

Answer: 3 groups are made

10 sharpeners cost Rs. 50, price of 1 sharpener

Step 1: Price of 10 sharpeners = Rs. 50.

Step 2: 50 divided by 10 = 5.

Answer: Rs. 5 for 1 sharpener

Learning Task (page 80) — Division

1. Divide 8 mangoes equally in 2 plates

Step 1: 8 divided by 2 = 4.

Answer: 4 mangoes on each plate

2. Divide 10 books equally among 2 students

Step 1: 10 divided by 2 = 5.

Answer: 5 books for each student

Review Exercise 1 (page 81) — Tick the correct option

a) Complete the table of 5: 5, 10, 15, ___ (options: 21, 25, 20)

- 21
- 25
- ✓ 20

b) Complete the table of 10: 10, 20, 30, ___ (options: 41, 40, 50)

- 41
- ✓ 40
- 50

c) 5 + 5 + 5 = ___ (options: 5 times 3, 3 times 5, 3 times 15). NOTE: 5 times 3 and 3 times 5 both equal 15; the book's grouping (3 groups of 5) is 3 times 5.

- 5 times 3
- ✓ 3 times 5

3 times 15

d) 2 x 10 = ___ (options: 22, 12, 20)

22

12

✓ 20

e) 7 x 2 = ___ (options: 7, 2, 14)

7

2

✓ 14

f) 10 divided by 5 = ___ (options: 5, 10, 2)

5

10

✓ 2

Review Exercise 2 (page 82) — Solve (flowers)

Total flowers: 4 bunches of 4 flowers

Step 1: Count the first bunch: 4 flowers.

Step 2: Count the second bunch: 4 flowers.

Step 3: Count the third bunch: 4 flowers.

Step 4: Count the fourth bunch: 4 flowers.

Step 5: 4 + 4 + 4 + 4 = 16.

Step 6: 4 times 4 = 16.

Answer: 4 x 4 = 16 flowers in total

Review Exercise 3 (page 82) — Solve the following

2 x 2

Step 1: 2 times 2 is 4.

Answer: 4

8 x 2

Step 1: 8 times 2 is 16.

Answer: 16

5 x 5

Step 1: 5 times 5 is 25.

Answer: 25

7 x 2

Step 1: 7 times 2 is 14.

Answer: 14

10 x 3

Step 1: 10 times 3 is 30.

Answer: 30

5 x 8

Step 1: 5 times 8 is 40.

Answer: 40

Review Exercise 4 (page 82) — Solve (order of division matters)

4 divided by 2 is NOT equal to 2 divided by 4

Step 1: 4 divided by 2 = 2.

Step 2: 2 divided by 4 is not 2.

Answer: 4 / 2 is not equal to 2 / 4

8 divided by 2 is NOT equal to 2 divided by 8

Step 1: 8 divided by 2 = 4.

Step 2: 2 divided by 8 is not 4.

Answer: 8 / 2 is not equal to 2 / 8

10 divided by 5 is NOT equal to 5 divided by 10

Step 1: 10 divided by 5 = 2.

Step 2: 5 divided by 10 is not 2.

Answer: 10 / 5 is not equal to 5 / 10

15 divided by 5 is NOT equal to 5 divided by 15

Step 1: 15 divided by 5 = 3.

Step 2: 5 divided by 15 is not 3.

Answer: 15 / 5 is not equal to 5 / 15

Review Exercise 5-6 (page 82)

5. 6 water bottles in a packet, 5 packets

Step 1: Bottles in one packet = 6.

Step 2: Number of packets = 5.

Step 3: 6 x 5 = 30.

Answer: 30 water bottles in 5 packets

6. 5 apples in a box, boxes needed for 30 apples

Step 1: Apples in one box = 5.

Step 2: 30 divided by 5 = 6.

Answer: 6 boxes are needed for 30 apples

Review Exercise 7-9 (page 83)

7. 18 students sit in 3 rows

Step 1: Total students = 18.

Step 2: Number of rows = 3.

Step 3: 18 divided by 3 = 6.

Answer: 6 students in each row

8. One water bottle costs Rs. 20, cost of 5 bottles

Step 1: Cost of 1 bottle = Rs. 20.

Step 2: 20 x 5 = 100.

Answer: Rs. 100 for 5 water bottles

9. 12 apples in a box, apples in 3 boxes

Step 1: Apples in 1 box = 12.

Step 2: 12 x 3 = 36.

Answer: 36 apples in 3 boxes

Unit 4 — Fractions (pages 84 to 93)

Students' learning outcomes (page 84): After completing this unit, students will be able to recognize, name and write fractions: half (1/2), quarter (1/4), two-quarter (2/4) and three-quarter (3/4) of a length, shape, set of objects or quantity using pictorial representations.

Fractions (page 85) — Half

Each part of a whole is called a fraction. A fraction is a part of a whole. A whole can be divided into 2 equal parts or 4 equal parts. One part out of two equal parts of a whole is half. We express one half as 1/2 and read it as 'one over two' or 'one by two'.

One Quarter (page 86)

One part out of four equal parts of a whole is one quarter. We express one quarter as 1/4 and read it as 'one over four' or 'one by four'.

Learning Task 1 (page 86) — Tick the shapes having two equal parts

Tick: the circle divided into two equal halves, the rectangle divided into two equal halves, the triangle divided into two equal halves, and the diamond divided into two equal halves. Do not tick the shapes whose two parts are unequal.

Learning Task 2 (page 86) — Colour half of each shape

Colouring task: colour one of the two equal parts of each shape (the first bar is already coloured as an example). No written answer is needed.

Learning Task 3 (page 87) — Tick the shapes having four equal parts

Tick: the bar divided into 4 equal parts, the circle divided into 4 equal quarters, the rectangle divided into 4 equal triangles, and the bar divided into 4 equal strips. Do not tick the circle or bar whose parts are unequal, the undivided triangle, or the circle divided into 3 parts.

Learning Task 4 (page 87) — Colour one quarter of each shape

Colouring task: colour one of the four equal parts of each shape (the first bar is already coloured as an example). No written answer is needed.

Learning Task 5 (page 88) — Write the fraction in symbols for the coloured parts

Shape — fraction of the coloured part

Circle, 1 of 2 parts coloured (given in the book) — 1/2

Bar, 1 of 4 parts coloured — 1/4

Rectangle cut diagonally, 1 of 2 parts coloured — 1/2

Rectangle of 4 triangles, 1 coloured — 1/4

Bar of 4 strips, 1 coloured — 1/4

Circle of 4 quarters, 1 coloured — 1/4

Small rectangle of 2 parts, 1 coloured — 1/2

Two-Quarters and Three-Quarters (page 89)

Two parts out of four equal parts of a whole is two-quarter. We express two-quarter as 2/4 and read it as 'two over four' or 'two by four'. Three parts out of four equal parts of a whole is three-quarter. We express three-quarter as 3/4 and read it as 'three over four' or 'three by four'.

Learning Task 1 (page 90) — Colour two quarters of each shape

Colouring task: colour two of the four equal parts of each shape. No written answer is needed.

Learning Task 2 (page 90) — Colour three quarters of each shape

Colouring task: colour three of the four equal parts of each shape. No written answer is needed.

Learning Task 3 (page 91) — Write the fraction for the coloured parts

Shape — fraction of the coloured part

Bar, 2 of 4 parts coloured (given in the book) — 2/4

Bar, 3 of 4 parts coloured — 3/4

Circle, 3 of 4 quarters coloured — 3/4

Bar of 4 strips, 2 coloured — 2/4

Diamond of 4 triangles, 3 coloured — 3/4

Triangle, 1 of 2 halves coloured — 1/2

Bar of 4 strips, 1 coloured — 1/4

Learning Task 4 (page 91) — Match the columns

Fraction name — matching picture

Half (1/2) — The shape of 2 parts with 1 coloured

One-Quarter (1/4) — The shape of 4 parts with 1 coloured

Two-Quarter (2/4) — The shape of 4 parts with 2 coloured

Three-Quarter (3/4) — The shape of 4 parts with 3 coloured

Review Exercise 1 (page 92) — Choose the correct answer

i) The bar has 2 of 4 parts coloured. The coloured part represents (options: 1/2, 2/4, 1/4, 3/4)

- 1/2
- ✓ 2/4
- 1/4
- 3/4

ii) Three quarter is (options: 3/4, 1/4, 2/4, 1/2)

- ✓ 3/4
- 1/4
- 2/4
- 1/2

iii) One part out of two equal parts of a whole is (options: one quarter,

two-quarters, three-quarters, half)

one quarter

two-quarters

three-quarters

✓ half

Q: iv) The circle has 3 of 4 quarters coloured. What fraction is it? NOTE (book error, flagged): none of the printed options (quarter, half, two-quarters, half) is correct.

A: Three-quarters (3/4). No printed option is ticked because the correct answer is not among the options.

Review Exercise 2 (page 93) — Colour half of the shapes

Colouring task: colour one of the two equal parts of each shape (circle, rectangle, diagonal rectangle). No written answer is needed.

Review Exercise 3 (page 93) — Colour three-quarters of the shapes

Colouring task: colour three of the four equal parts of each shape (4-part rectangle, 4-strip bar, 4-triangle diamond). No written answer is needed.

Review Exercise 4 (page 93) — Write the fractions in symbols for the coloured parts

Shape — fraction of the coloured part

Bar of 4 parts, 1 coloured pink — 1/4

Circle of 4 quarters, 3 coloured pink — 3/4

Triangle of 2 halves, 1 coloured pink — 1/2

Review Exercise 5 (page 93) — Match the Columns

Name — fraction

Half — 1/2

One quarters — 1/4

Three quarters — 3/4

Two quarters — 2/4

Unit 5 — Money (pages 94 to 103)

Students' learning outcomes (page 94): After completing this unit, students will be able to identify Pakistani coins (Rs. 1, 2, 5 and 10) and notes (Rs. 10, 20, 50, 100 and 500), and solve money problems involving addition and subtraction of Pakistani currency.

Pakistani Coins (page 95) — Look and recognize

Coin — value

Coin with 1 — Rupee 1

Coin with 2 — Rupees 2

Coin with 5 — Rupees 5

Coin with 10 (Faisal Mosque) — Rupees 10

Pakistani Currency Notes (page 96) — Look and recognize

Note — value

Green note, 10 — Rs. 10

Orange note, 20 — Rs. 20

Purple note, 50 — Rs. 50

Red note, 100 — Rs. 100

Note, 500 — Rs. 500

Learning Task (page 97) — Match the coins and notes to the values of given objects

Money — object with that price

Rs. 20 note — Pencil, Rs. 20

Rs. 50 note — Cupcake, Rs. 50

Rs. 10 coin — Sharpener, Rs. 10

Rs. 100 note — Ice cream, Rs. 100

Rs. 5 coin — Eraser, Rs. 5

Addition of Money (page 98)

Ali bought a toy car (Rs. 48) and a bat (Rs. 50)

Step 1: Price of car = Rs. 48.

Step 2: Price of bat = Rs. 50.

Step 3: 48 + 50 = 98.

Answer: Ali spent Rs. 98

Zarina bought a bat (Rs. 50) and a ball (Rs. 35)

Step 1: Price of bat = Rs. 50.

Step 2: Price of ball = Rs. 35.

Step 3: 50 + 35 = 85.

Answer: Zarina spent Rs. 85

Learning Task (page 99) — Read the prices and add

Prices shown: Mango Rs. 40, Apple Rs. 32, Milk Rs. 35, Cake Rs. 30, Doll Rs. 45.

a) Mango + Milk

Step 1: Price of a mango = Rs. 40.

Step 2: Price of milk = Rs. 35.

Step 3: 40 + 35 = 75.

Answer: Total price Rs. 75

b) Cake + Apple

Step 1: Price of a cake = Rs. 30.

Step 2: Price of an apple = Rs. 32.

Step 3: 30 + 32 = 62.

Answer: Total price Rs. 62

c) Doll + Apple

Step 1: Price of a doll = Rs. 45.

Step 2: Price of an apple = Rs. 32.

Step 3: 45 + 32 = 77.

Answer: Total price Rs. 77

d) Mango + Doll

Step 1: Price of a mango = Rs. 40.

Step 2: Price of a doll = Rs. 45.

Step 3: 40 + 45 = 85.

Answer: Total price Rs. 85

Subtraction of Money (page 100)

Marwa had Rs. 80, bought pizza for Rs. 40

Step 1: Marwa had Rs. 80.

Step 2: Price of pizza = Rs. 40.

Step 3: 80 - 40 = 40.

Answer: Rs. 40 left

Learning Task 1 (page 100) — Ali had Rs. 85, bought a cap for Rs. 40

85 - 40

Step 1: Ali had Rs. 85.

Step 2: Price of cap = Rs. 40.

Step 3: 85 - 40 = 45.

Answer: Rs. 45 left

Learning Task 2 (page 100) — Arwa had Rs. 78, bought a notebook for Rs. 53

78 - 53

Step 1: Arwa had Rs. 78.

Step 2: Price of notebook = Rs. 53.

Step 3: 78 - 53 = 25.

Answer: Rs. 25 left

Learning Task 3 (page 100) — Ali had Rs. 50, gave Rs. 30 to his sister

50 - 30

Step 1: Ali had Rs. 50.

Step 2: Gave to sister = Rs. 30.

Step 3: 50 - 30 = 20.

Answer: Rs. 20 left

Review Exercise 1-5 (page 101) — Choose the correct option

1) A 5-rupee coin + a 5-rupee coin is equal to (options: Rs. 5, Rs. 10, Rs. 15, Rs. 20)

- Rs. 5
- ✓ Rs. 10
- Rs. 15
- Rs. 20

2) Five, 2-rupee coins make (options: Rs. 10, Rs. 8, Rs. 4, Rs. 6)

- ✓ Rs. 10
- Rs. 8
- Rs. 4
- Rs. 6

3) A Rs. 10 note + a Rs. 20 note = (options: Rs. 10, Rs. 20, Rs. 30, Rs. 40)

- Rs. 10
- Rs. 20
- ✓ Rs. 30
- Rs. 40

4) Rs. 45 + Rs. 15 = (options: Rs. 60, Rs. 70, Rs. 75, Rs. 40)

- ✓ Rs. 60
- Rs. 70
- Rs. 75
- Rs. 40

5) Rs. 50 - Rs. 20 = (options: Rs. 50, Rs. 20, Rs. 30, Rs. 15)

- Rs. 50
- Rs. 20
- ✓

Rs. 30

Rs. 15

Review Exercise 6 (page 102) — Write the value of coins and notes

Money shown — value

2-rupee coin — Rs. 2

5-rupee coin — Rs. 5

10-rupee coin — Rs. 10

Rs. 20 note — Rs. 20

Rs. 50 note — Rs. 50

Rs. 100 note — Rs. 100

Review Exercise 7 (page 102) — Match the columns A and B

Money (Column A) — value (Column B)

5-rupee coin — Rs. 5

Rs. 100 note — Rs. 100

Rs. 50 note — Rs. 50

Rs. 20 note — Rs. 20

Review Exercise 8 (page 102) — Asiya had Rs. 50, bought a ball for Rs. 37

50 - 37

Step 1: Asiya had money = Rs. 50.

Step 2: Bought a ball for = Rs. 37.

Step 3: 50 - 37 = 13.

Answer: Rs. 13 left with Asiya

Review Exercise 9 (page 103) — Sohail bought milk and bread

Milk Rs. 25 + bread Rs. 30

Step 1: Price of milk = Rs. 25.

Step 2: Price of bread = Rs. 30.

Step 3: 25 + 30 = 55.

Answer: Sohail spent Rs. 55 altogether

Review Exercise 10 (page 103) — Balls in a basket

34 red balls + 41 green balls

Step 1: Red balls = 34.

Step 2: Green balls = 41.

Step 3: 34 + 41 = 75.

Answer: 75 balls in all

Review Exercise 11 (page 103) — Rifat and the shopkeeper

Rifat gave Rs. 50 for Rs. 30 ice cream

Step 1: Total money = Rs. 50.

Step 2: Cost of ice cream = Rs. 30.

Step 3: 50 - 30 = 20.

Answer: Rifat will get back Rs. 20

Unit 6 — Measurement (pages 104 to 116)

Students' learning outcomes (page 104): After completing this unit, students will be able to measure and compare the length of only two objects using non-standard units; use mathematical language to compare the mass of two or more objects; measure and compare the mass of objects using non-standard units; measure and compare the capacity of objects using non-standard units; read and write temperature to the nearest appropriate unit in degrees centigrade (C) using pictorial representation and relating the temperature scale to the number line.

Length (page 105) — Long, Longer, Longest

Length tells us how long something is from one end to the other. Long: car. Longer: bus. Longest: truck. Long: short pencil. Longer: middle pencil. Longest: longest pencil.

Learning Task 1 (page 105) — Tick the longest

Q: Snakes: which is the longest?

A: The third (rightmost) snake is the longest — tick its box.

Q: Wood pieces: which is the longest?

A: The middle wood piece is the longest — tick its box.

Learning Task 2 (page 106) — Colour the Long (Red) and the Longest (Green)

Q: a) Fish: which is long, which is longest?

A: Colour the rightmost (middle-sized) fish Red (long) and the leftmost (largest) fish Green (longest). The small middle fish is neither.

Q: b) Carrots: which is long, which is longest?

A: Colour the rightmost carrot Red (long) and the middle carrot Green (longest). The small left carrot is neither.

Q: c) Ropes: which is long, which is longest?

A: Colour the middle rope Red (long) and the rightmost rope Green (longest). The small left rope is neither.

Height (page 107) — Short, Shorter, Shortest

Height is the measurement of something or someone from top to bottom. Short pencil: the tallest of the three pencils shown. Shorter pencil: the middle pencil. Shortest pencil: the smallest pencil.

Learning Task 1 (page 107) — Tick the shortest

Q: a) Combs: which is the shortest?

A: The middle (smallest) comb is the shortest — tick its box.

Q: b) Arrows: which is the shortest?

A: The middle (smallest) arrow is the shortest — tick its box.

Tall, Taller, Tallest (page 108)

Tall: the smallest candle. Taller: the middle candle. Tallest: the largest candle.

Learning Task 1 (page 108) — Tick the tallest bottle

Q: Which bottle is the tallest?

A: The middle bottle is the tallest — tick its box.

Learning Task 2 (page 108) — Tick the shortest boy

Q: Which boy is the shortest?

A: The leftmost (smallest) boy is the shortest — tick its box.

Mass (page 109) — Heavy, Heavier, Heaviest; Light, Lighter,

Lightest

Mass refers to the quantity of matter in an object. Heavy, heavier, heaviest and light, lighter, lightest are the words used to compare the mass of objects.

Learning Task (page 110) — Mass

Q: 1. Tick the heaviest object and cross the lightest one (chick, bear, rooster).

A: Tick the bear (heaviest) and cross the chick (lightest).

Q: 2. Colour the light (Red) and lightest (Green) (comb, bucket, leaf).

A: Colour the comb Red (light) and the leaf Green (lightest). The bucket is the heaviest of the three.

Q: 3. Tick the heaviest object (rooster, chick, egg).

A: Tick the rooster (heaviest).

Q: 4. Tick the lightest object (apple, watermelon, strawberries).

A: Tick the strawberries (lightest).

Capacity (page 111) — More and less liquid

Capacity refers to the maximum amount or space something can contain. The jug contains more water than the glass. The kettle contains more tea than the cup. The bottle contains less water than the cane (bucket). The mug contains less milk than the thermos.

Learning Task 1 (page 111) — Tick the container that holds more liquid

Q: Bowl or bottle: which holds more?

A: The bottle holds more liquid — tick it.

Q: Small bottle or big water can: which holds more?

A: The big water can holds more liquid — tick it.

Learning Task (page 112) — Holds more, holds less, half jug

Q: Cup or kettle: which holds more?

A: The kettle holds more liquid — tick it.

Q: Milk carton or cooking oil can: which holds more?

A: The cooking oil can holds more liquid — tick it.

Q: Small oil bottle or big oil bottle: which holds more?

A: The big oil bottle holds more liquid — tick it.

Q: 2. Tick the container that holds less liquid (small milk carton or 1 kg milk carton; cup or kettle; jar or glass).

A: Tick the small milk carton, the cup, and the glass — each holds less than its partner.

Full, Half, Empty: the first glass is full of water, the second glass is half, the third glass is empty.

Q: 3. Tick the half jug.

A: The middle jug, filled halfway, is the half jug — tick it.

Temperature (page 113)

Temperature tells us how hot or cold something is. We measure it in degrees centigrade (C). On the number line with the temperature scale: 0 C is a cold day and 35 C is a hot day.

Learning Task (page 114) — Read and write the temperature on the thermometer

Q: Temperature of Thermometer A is ___ (read approximately from the scale).

A: About 35 C. NOTE: thermometer readings are read approximately from the printed scale.

Q: Temperature of Thermometer B (labelled Thermometer 2 in the book) is ___.

A: About 20 C, read approximately from the printed scale.

Review Exercise 1 (page 115) — Tick the longest object

Q: Which carrot is the longest?

A: The middle carrot is the longest — tick its box.

Review Exercise 2 (page 115) — Tick the shortest object

Q: Which vehicle is the shortest?

A: The car (leftmost) is the shortest — tick its box.

Review Exercise 3 (page 115) — Tick the shortest tree

Q: Which tree is the shortest?

A: The leftmost (smallest) tree is the shortest — tick its box.

Review Exercise 4 (page 115) — Write 'A' for high, 'B' for higher, 'C' for highest

Q: Three buildings, left to right: which gets A, B, C?

A: Left (smallest) building: A (high). Middle building: B (higher). Right (tallest) building: C (highest).

Review Exercise 5 (page 116) — Tick the heavy and cross the heaviest object (stones)

Q: Three stones, left to right: which is ticked, which is crossed?

A: Tick the middle stone (heavy) and cross the leftmost, largest stone (heaviest). The smallest stone is neither.

Review Exercise 6 (page 116) — Write 'F' for full, 'H' for half, 'E' for empty glass

Q: Three glasses, left to right: F, H or E?

A: Left glass: F (full). Middle glass: H (half). Right glass: E (empty).

Review Exercise 7 (page 116) — Read and write the temperature on the thermometer

Q: Temperature of Thermometer A is ___ (read approximately from the scale).

A: About 25 C, read approximately from the printed scale.

Q: Temperature of Thermometer B (labelled Thermometer 2 in the book) is ___.

A: About 5 C, read approximately from the printed scale.

Unit 7 — Time and Date (pages 117 to 128)

Students' learning outcomes (page 117): After completing this unit, students will be able to read and write time in hours (o'clock) from an analogue clock and a digital clock; show time in an hour on an analogue clock; name the days of the week and the months of the solar and Islamic year.

Time (page 118) — Analogue and digital clocks

An analogue clock has two hands. The hour hand is shorter than the

minute hand. In the example the hour hand is at 3 and the minute hand is at 12, so the time is 3 o'clock. A digital clock has no hands; it tells the time using digits. In the example it shows 8:00: the first part tells the hours and the second part tells the minutes.

Learning Task 1 (page 119) — Read and tell the time

Q: First clock (hour hand at 2, minute hand at 12).

A: 2 o'clock.

Q: Second clock (hour hand at 4, minute hand at 12).

A: 4 o'clock.

Q: Third clock (hour hand at 7, minute hand at 12).

A: 7 o'clock.

Learning Task 2 (page 119) — Read and write the time

Q: First clock (hour hand at 6, minute hand at 12).

A: 6 o'clock.

Q: Second clock (hour hand at 1, minute hand at 12).

A: 1 o'clock.

Q: Third clock (hour hand at 8, minute hand at 12).

A: 8 o'clock.

Learning Task 3 (page 119) — Draw the long and short hands to show the given time

Q: Clock for 2:00: where do the hands go?

A: Short (hour) hand at 2 and long (minute) hand at 12.

Q: Clock for 7:00: where do the hands go?

A: Short (hour) hand at 7 and long (minute) hand at 12.

Q: Clock for 11:00: where do the hands go?

A: Short (hour) hand at 11 and long (minute) hand at 12.

Learning Task 4 (page 120) — What time is it? (digital clocks)

Q: Digital clock showing 3:00.

A: 3 O'Clock.

Q: Digital clock showing 9:00.

A: 9 O'Clock.

Q: Digital clock showing 6:00.

A: 6 O'Clock.

Q: Digital clock showing 8:00.

A: 8 O'Clock.

Q: Digital clock showing 11:00.

A: 11 O'Clock.

Q: Digital clock showing 1:00.

A: 1 O'Clock.

Learning Task 5 (page 120) — Write the given time in a digital clock

Q: 5 O'Clock in digital form.

A: 5:00

Q: 8 O'Clock in digital form.

A: 8:00

Q: 11 O'Clock in digital form.

A: 11:00

Learning Task 6 (page 120) — Match the columns (analogue to digital)

Analogue clock — digital time

Hour hand at 3, minute hand at 12 — 3:00

Hour hand at 9, minute hand at 12 — 9:00

Hour hand at 12, minute hand at 12 — 12:00

Hour hand at 6, minute hand at 12 — 6:00

Key point (page 120): Hour is denoted by 'h' and minute is denoted by 'm'.

Days of the Week (page 121)

Day — position in the week (book's order)

Monday — 1st day of the week

Tuesday — 2nd day of the week

Wednesday — 3rd day of the week

Thursday — 4th day of the week

Friday — 5th day of the week

Saturday — 6th day of the week

Sunday — 7th day of the week

Learning Task 1 (page 121) — Match the columns

Day — position

Monday — 1st day

Tuesday — 2nd day

Wednesday — 3rd day

Thursday — 4th day

Friday — 5th day

Saturday — 6th day

Sunday — 7th day

Learning Task 2 (page 122) — Put the days of the week in order

In order: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday.

Months of the Year (page 122) — Solar year

Month — position

January — 1st

February — 2nd

March — 3rd

April — 4th

May — 5th

June — 6th

July — 7th

August — 8th

September — 9th

October — 10th

November — 11th

December — 12th

Learning Task (page 123) — Write the correct order of the months of the year

In order: January, February, March, April, May, June, July, August, September, October, November, December.

Key point (page 123): 1 day = 24 hours. 1 week = 7 days. 1 month = 30 days.

Calendar 2025 (page 124) — Questions on the solar calendar

Q: Name the day on 4th February.

A: Tuesday (4 February 2025 was a Tuesday).

Q: Name the day on 3rd March.

A: Monday (3 March 2025 was a Monday).

Q: Find 7th June in the solar calendar (name the day).

A: Saturday (7 June 2025 was a Saturday).

Q: Find 14th August in the solar calendar (name the day).

A: Thursday (14 August 2025 was a Thursday).

Q: How many Sundays are there in the month of May?

A: 4 Sundays: 4th, 11th, 18th and 25th May 2025.

Names of Islamic Months (page 125)

In order: Muharram (1st), Safar (2nd), Rabi-ul-Awal (3rd), Rabi-ul-Sani (4th), Jumadi-ul-Awal (5th), Jumadi-ul-Sani (6th), Rajab (7th), Sha'baan (8th), Ramadan (9th), Shawwal (10th), Dhu-al-Qudah (11th), Dhu-al-Hijah (12th).

Activity (page 125) — Islamic months

Memorizing activity: memorize the names of the Islamic months and repeat them to the teacher. No written answer is needed.

Review Exercise 1 (page 126) — Choose the correct option

i) Number of days in a week (options: 6, 7, 8)

6



7

8

ii) First day of the week is (options: Friday, Saturday, Monday)

Friday

Saturday

✓ Monday

iii) Second month of the Solar year is (options: January, February, March)

January

✓ February

March

iv) Analogue Clock has the numbers (options: 1 to 6, 1 to 9, 1 to 12)

1 to 6

1 to 9

✓ 1 to 12

v) The time in the given clock is (hour hand at 3, minute hand at 12; options: 3 O'clock, 12 O'clock, 2 O'clock)

✓ 3 O'clock

12 O'clock

2 O'clock

Review Exercise 2 (page 127) — Read and write the time

Q: First analogue clock (hour hand at 9, minute hand at 12).

A: 9 O'clock.

Q: Second analogue clock (hour hand just past 12, minute hand at 5).

NOTE: this is the unit's only clock that is not on the hour.

A: 12:25 (25 minutes past 12).

Q: Digital clock showing 5:00.

A: 5 O'Clock.

Q: Digital clock showing 9:00.

A: 9 O'Clock.

Review Exercise 3 (page 127) — Draw hands to show the given time

Q: Clock for 9 O'Clock: where do the hands go?

A: Short (hour) hand at 9 and long (minute) hand at 12.

Q: Clock for 3 O'Clock: where do the hands go?

A: Short (hour) hand at 3 and long (minute) hand at 12.

Q: Clock for 6 O'Clock: where do the hands go?

A: Short (hour) hand at 6 and long (minute) hand at 12.

Review Exercise 4 (page 127) — Match the Columns

Day — position

Monday — 1st

Tuesday — 2nd

Wednesday — 3rd

Thursday — 4th

Friday — 5th

Saturday — 6th

Sunday — 7th

Review Exercise 2 (page 128) — Read the time from the analogue clock and write in the boxes

Q: First clock (hour hand at 1, minute hand at 6). NOTE: this clock is not

on the hour.

A: 1:30 (half past 1).

Q: Second clock (hour hand at 3, minute hand at 12).

A: 3 O'clock.

Q: Third clock (hour hand at 9, minute hand at 12).

A: 9 O'clock.

Q: Fourth clock (hour hand at 6, minute hand at 12).

A: 6 O'clock.

Q: Fifth clock (hour hand at 9, minute hand at 12).

A: 9 O'clock.

Q: Sixth clock (hour hand at 10, minute hand at 12).

A: 10 O'clock.

Unit 8 — Geometry and Patterns (pages 129 to 147)

Students' learning outcomes (page 129): After completing this unit, students will be able to recognize and identify 2-D shapes (rectangle, square, circle and triangle) with respect to their characteristics (sides and corners); recognize and identify 3-D shapes (cube, cuboid, cone, cylinder, sphere) with respect to their characteristics; describe the position, movement and direction of an object using positional language (inside, outside, above, below, over, under, near, far, before, after, straight, forward, backward, right and left); identify and extend the next shape in a pattern with 2 or 3 orientations; identify and extend patterns using 2-D

and 3-D shapes and pictorial illustrations.

2-D Shapes (pages 130-131)

Square: it has 4 corners and 4 sides. All sides are equal in length. Rectangle: it has 4 corners and 4 sides. Opposite sides are equal in length. Triangle: it has 3 corners and 3 sides. Circle: it has a curved side and no corners.

Learning Task 1 (page 132) — Write the number of sides and corners

Shape — sides and corners

Rectangle — Sides: 4, Corners: 4

Triangle — Sides: 3, Corners: 3

Square — Sides: 4, Corners: 4

Circle — Sides: 1 curved side, Corners: 0

Learning Task 2 (page 132) — Colour the squares (Red), circles (Yellow), rectangles (Green) and triangles (Blue)

Colouring task for the house picture: colour the 2 triangles Blue (the roof triangle and the inverted triangle), the 3 circles Yellow (two round windows and the round door window), the 3 squares Red (the three small square windows), and all 8 rectangles Green (roof rectangles, door and bottom windows). No written answer is needed.

Learning Task 3 (page 133) — Match the object with their similar shape

Object — similar shape

Envelope — Rectangle

Mirror (square frame) — Square

Clock (round) — Circle

Road sign (triangular) — Triangle

3-D Shapes (pages 133-134)

Cube: a solid 3-dimensional shape. It has six square faces and 8 corners. Examples: box, dice, purple cube. Cuboid: a rectangular 3-dimensional shape. It has 6 rectangular faces and 8 corners. Sphere: a round ball 3-D shape; every part of its outside is the same distance from the middle. Cone: it has 1 flat face and 1 curved face, pointed towards the top. Cylinder: a solid 3-D shape; it has 2 flat faces and 1 curved face.

Learning Task (page 135) — Match the shapes with their similar shapes

Object — 3-D shape

Eraser — Cuboid

Ball — Sphere

Gift box — Cube

Drink can — Cylinder

Party hat — Cone

Patterns (page 135)

A pattern is the way in which objects are arranged according to a specific rule. Examples on the page: cricket bat and ball repeat (bat, ball, bat, ball...); red square, blue rectangle and yellow circle repeat (square,

rectangle, circle, square, rectangle, circle...).

Learning Task 1 (page 136) — Colour the shape to complete the pattern

Q: Row 1 (red triangle, yellow circle repeating): what colours complete the last two shapes?

A: Colour the triangle Red and the circle Yellow.

Q: Row 2 (blue square, green rectangle, red circle repeating): what colours complete the last two shapes?

A: Colour the rectangle Green and the circle Red.

Learning Task 2 (page 136) — Draw and colour one shape that comes next

Q: Row 1 (black square, blue rectangle repeating): what comes next?

A: A blue rectangle.

Q: Row 2 (blue diamond, yellow balloon repeating): what comes next?

A: A yellow balloon.

Q: Row 3 (square, tall rectangle, circle repeating): what comes next?

A: A circle.

Learning Task 3 (page 137) — What comes next to complete the pattern?

Q: a) Yellow circle, red triangle, pink square repeating: what two shapes come next?

A: A red triangle, then a pink square.

Q: b) Green rectangle, yellow circle, red triangle repeating: what two shapes come next?

A: A yellow circle, then a red triangle.

Q: c) Blue, red, purple triangles repeating: what two shapes come next?

A: A red triangle, then a purple triangle.

Positions — Inside, Outside (page 137)

The lion is inside the cage. The lion is outside the cage.

Try Yourself (page 138) — Inside and outside

Q: 1. Circle the bird inside the cage.

A: Circle the bird that is inside the cage.

Q: 2. Circle the girl outside the box.

A: Circle the girl that is outside the box.

Q: 3. Circle the apples inside the shopper (bag).

A: Circle the apples that are inside the shopper.

Q: 4. Circle the eggs outside the nest.

A: Circle the eggs that are outside the nest.

Above, Below (page 138)

The book is above the table. The cat is below the table.

Learning Task 1 (page 139) — Tick the object above the table and cross the object below the table

Tick the object drawn above the table and cross the object drawn below the table.

Learning Task 2 (page 139) — Tick the bird above the tree and cross the bird below the tree

Tick the bird drawn above the tree and cross the bird drawn below the tree.

Over, Under (page 140)

The boy is over the bridge. The duck is under the bridge.

Learning Task (page 140) — Fill in the blank with over and under

Boy is ___ the table.

Missing word: over

Girl is ___ the table.

Missing word: under

Far and Near (page 141)

If an object is located at a short distance it is near, and if it is located at a long distance it is far away. The boy is near the house (left picture). The boy is far from the house (right picture).

Learning Task (page 141) — Write (F) for far and (N) for near object

Q: a) Boy far from the school building.

A: F (the boy is far from the school).

Q: b) Boy close to the school building.

A: N (the boy is near the school).

After and Before (page 142)

The cat is after the rabbit. The rabbit is after the dog. The dog is before the rabbit.

Learning Task 1 (page 142) — Encircle the bird that is after the duck

Q: Chick, duck, parrot in a row: which bird is after the duck?

A: The parrot (rightmost) is after the duck — encircle it.

Learning Task 2 (page 142) — Encircle the bicycle that is before the car

Q: Red bicycle, car, black bicycle in a row: which bicycle is before the car?

A: The red bicycle (leftmost) is before the car — encircle it.

Learning Task 3 (page 142) — Tick the aeroplane after the red one

Q: Yellow, red and pink aeroplanes in a row: which aeroplane is after the red one?

A: The pink aeroplane (rightmost) is after the red one — tick it.

Right and Left (page 143)

Amna is going to the left hand side. Ali is going to the right hand side.

Learning Task (page 143) — Tick the Right hand side and cross the left hand side

Q: Bicycle on the left of the picture, car on the right: which is ticked, which is crossed?

A: Tick the car (right hand side) and cross the bicycle (left hand side).

Review Exercise 1 (page 144) — Choose the correct option

i) The number of corners of a triangle is (options: 3, 4, 5)

✓ 3

4

5

ii) The number of sides of a rectangle is (options: 3, 4, 8)

3

✓

4

8

iii) A circle has corners (options: 0, 1, 3)

✓ 0

1

3

iv) A blackboard has a similar shape to the (options: triangle, circle, rectangle)

triangle

circle

✓ rectangle

v) Four sides of the shape are equal (options: square, rectangle, triangle)

✓ square

rectangle

triangle

Review Exercise 2 (page 145) — Write the shape of the following objects

Object — shape

Mirror (rectangular frame) — Rectangle

Road sign — Triangle

Clock (square frame) — Square

Tyre — Circle

Review Exercise 3 (page 145) — Write numbers of corners and sides of the shapes

Shape — corners and sides

Rectangle (green) — Corners: 4, Sides: 4

Circle (purple) — Corners: 0, Sides: 1 curved side

Triangle (brown) — Corners: 3, Sides: 3

Square (blue) — Corners: 4, Sides: 4

Review Exercise 4 (page 145) — Complete the patterns

Q: Row 1 (triangle, rectangle, circle repeating): what three shapes complete the row?

A: A circle, then a triangle, then a rectangle.

Q: Row 2 (square, circle, rectangle repeating): what three shapes complete the row?

A: A rectangle, then a square, then a circle.

Review Exercise 5 (page 146) — Verbally tell the position of the following

Q: Ball and table (two pictures).

A: First picture: the ball is under the table. Second picture: the ball is on (above) the table.

Q: Cat and box (two pictures).

A: First picture: the cat is inside the box. Second picture: the cat is outside (in front of) the box.

Q: Man and woman on a bench (two pictures).

A: First picture: the man and woman are far apart. Second picture: the man and woman are near (close together).

Q: Dogs and table.

A: The white dog is on the table and the black dog is under the table.

Review Exercise 6 (page 147) — Fill in the boxes (3-D shapes)

Shape — vertices, faces, edges

Cube (given in the book) — Vertices: 8, Faces: 6, Edges: 12

Cuboid — Vertices: 8, Faces: 6, Edges: 12

Sphere — Vertices: 0, Faces: 1 curved surface, Edges: 0

Cylinder — Vertices: 0, Faces: 3 (2 flat + 1 curved), Edges: 2

Cone — Vertices: 1, Faces: 2 (1 flat + 1 curved), Edges: 1

Unit 9 — Statistics and Graphs (pages 148 to 157)

Students' learning outcomes (page 148): After completing this unit, students will be able to read and interpret data using pictographs, block graphs and tally charts (including real-world problems); describe the likelihood that everyday events will occur, using mathematical language (impossible, less likely and more likely).

Identification of Pictograph (page 149)

There are 15 coloured pencils in total. Arranged by colour: there are 3 green pencils, 4 yellow pencils, 4 red pencils and 4 blue pencils. 3 + 4 + 4 + 4 = 15.

Bar Graph (page 150)

Reading the bar graph (number of students per subject): Urdu 10, English 30, Maths 20, Science 40, Islamiat 50. There are 50 students in the

Islamiat class — it is the greatest number of students. There are 10 students in the Urdu class — it is the lowest number of students. Total number of students: 10 + 30 + 20 + 40 + 50 = 150. Total number of subjects: 5 (Urdu, English, Maths, Science and Islamiat).

Tally Chart (page 151)

Class 1 students chose their favourite colours. Tally marks: Blue 4, Orange 4, Green 3, Black 10, Red 9. (One tally mark stands for one student.) There are 4 students who chose the blue colour, 4 who chose orange, and 3 who chose green. Key point: the marks in the tally column are equal to the number of observations in the data.

Probability / Likelihood (page 152)

Probability means how likely something is to happen. When you toss a coin there are two possible results: heads and tails. A mug has 5 pens: 4 blue pens and 1 red pen. Picking the blue pen is more likely (4 out of 5 pens are blue). Picking the red pen is less likely (only 1 out of 5 pens is red). Picking a yellow pen is impossible (there is no yellow pen in the mug).

Learning Task (page 153) — Likelihood of the balls in the box

The box has 10 balls in total: 4 green balls and 6 blue balls.

Q: More likely: which colour?

A: Blue ball (6 out of 10 balls are blue) — the book already fills this row.

Q: Less likely: which colour?

A: Green ball (only 4 out of 10 balls are green).

Q: Impossible: which colour?

A: Any colour not in the box, for example a red ball (there are no red balls

in the box).

Review Exercise 1 (page 154) — Read the pictograph and answer

Pictograph key: one square = 1 child. Monday 3, Tuesday 4, Wednesday 2, Thursday 6, Friday 1, Saturday 5.

Q: a) How many students are present on Saturday?

A: 5 students.

Q: b) On which day are 3 students present?

A: Monday.

Q: c) Name the day when one student is present.

A: Friday.

Q: d) How many students are present on Thursday?

A: 6 students.

Review Exercise 2 (page 155) — Toys sold on Monday at the toy shop

Bar graph: Car bar reaches 10, Elephant bar reaches 20, Balls bar reaches 50.

Q: a) How many elephants are sold?

A: 20 elephants.

Q: b) How many balls are sold?

A: 50 balls.

Review Exercise 3 (page 156) — Count the tally marks and write the numbers

Tally marks — number

III (given in the book) — 3

One gate of 5 — 5

One gate of 5 + III — 8

Two gates of 5 — 10

Two gates of 5 + II — 12

Two gates of 5 + IIII — 14

Three gates of 5 — 15

Three gates of 5 + II — 17

Three gates of 5 + IIII — 19

Four gates of 5 — 20

Review Exercise 4 (page 157) — Look at the picture and encircle the correct answer

The basket has 8 apples (6 in the top row, 2 in the bottom row) and 4 oranges, 12 fruits in total. There are no bananas.

i) How likely do you pick up an apple? (options: more likely, less likely, impossible)

✓ more likely

less likely

impossible

ii) How likely do you pick up an orange? (options: more likely, less likely, impossible)

more likely

✓ less likely

impossible

iii) How likely do you pick up a banana? (options: more likely, less likely, impossible)

more likely

less likely

✓ impossible

Glossary (Book page 158)

Ascending Order: numbers from smallest to largest. Addition: the process of calculating 2 or more numbers. Bar Graph: a graph which displays units of data with blocks. Backward: the process of counting downward. Currency: the system of money in general use. Coin: metal money is a coin. Cone: a 3-D figure. Circle: a 2-D figure, a round plain figure. Cube: a 3-D symmetrical shape. Descending Order: numbers from largest to smallest. Division: the process of being separated equally. Difference: the result of subtracting one number from another. Fraction: parts of a whole. Forward Counting: counting sequentially from a lower number to a higher number. Half: one of the two equal parts of a whole. Multiplication: the process of calculating the product of two or more numbers. Ordinal Number: a number defining the position of something. Ordering: placing objects into a particular arrangement. Quarter: each of 4 equal parts of a whole. Subtraction: the process of taking one number away from the other. Sum: the result after adding two or more numbers. Square: a 4-sided shape in which all sides are equal. Triangle: a plain figure with 3 sides and 3 corners.

Book pages 159-160 carry only 'Your valuable suggestions' feedback forms — no exercises, so there is nothing to solve there.

Judgement calls and book errors in Units 3-9 (kept flagged, not silently fixed): (1) Page 69, toy cars: the book's frame prints '4 + 4' but the

picture shows 16 cars (4 + 4 + 8) — counted total 16 used. (2) Page 81, Review 1(c): both '5 times 3' and '3 times 5' equal 15; the book's grouping (3 groups of 5) is 3 times 5. (3) Page 92, Review 1(iv): the circle shows 3/4 coloured but none of the printed options is three-quarters — answered as three-quarters with no option ticked. (4) Thermometer readings (pages 114, 116) are approximate readings from the printed scales, stated as 'about'. (5) Page 127, Review 2, second clock (12:25) and page 128, first clock (1:30) are the unit's only clocks that are not on the hour. (6) Page 147, sphere/cylinder/cone vertices-faces-edges are given at Grade 1 level (sphere: 0 vertices, 1 curved surface, 0 edges; cylinder: 0 vertices, 3 faces, 2 edges; cone: 1 vertex, 2 faces, 1 edge).